EE3001: Solid State Devices
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Assignment #7

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10}cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$3.2 \times 10^{19}cm^{-3}$, N_v=$1.8 \times 10^{19}cm^{-3}$, μ_n = $1350cm^2/V.s$, μ_p = $450cm^2/V.s$; Permittivity for vacuum: ϵ_0=8.85×10^{-14}F/cm, relative permittivity: $\epsilon_{r,Si}$=11.8, ϵ_{r,SiO_2}=3.9; pn diode dopants are completely ionized.

1. (2 points) At T=300 K, a p^+n junction with a long n region and reverse saturation current of $I_0 = 1\ \mu A$, is carrying a steady forward current $I_1 = 1$ mA for $t < 0$. At $t = 0$, the diode current is abruptly increased to $I_2 = 10$ mA. Calculate the diode voltage at $t = 0.5\ \mu s$ if $\tau_p = 1\ \mu s$.

2. (5 points) In switching applications, a diode has to be switched from forward to reverse bias condition and back. Assume that the diode under investigation is a p^+n junction diode with a long n region with a reverse saturation current of $I_0 = 1\ \mu A$ and is operating at 300 K. The switching diode is connected in series with a load resistor R=1 kΩ. The diode and the load resistor is driven by a voltage source that switches from +10 V to -5 V with a low enough switching frequency. The state of the voltage source is +10 V allowing a steady forward current $I_1 = 10$ mA through the diode for $t < 0$. At $t = 0$, the diode is abruptly reverse biased to -5 V. Now the stored minority carrier will gradually deplete; hence, will take some time to rearrange themselves according to the external bias condition. Given that $\tau_p = 1\ \mu s$.

   (a) At what time (known as storage delay time) will the stored minority charge become zero?

   (b) What is the final diode voltage and current?

   (c) If the voltage source switches between +5 V to -10 V, what would be the value of this time constant calculated in (a)?

   (d) If the diode under consideration corresponds to a p^+-n-n^+ structure with the quasi-neutral n-region length as 1 μm, find out t_{sd} for this device if the voltage source switches between +10 V to -5 V and between +5 V to -10 V.

3. (5 points) In a p^+-n-n^+ diode, the doping concentration of the n-region is 2×10^{15} cm^{-3}. If the critical field at avalanche breakdown is 0.2 MV/cm, answer the folowing.

(a) What will be the breakdown voltage of the diode if the width of the n-region is 10 μm? Sketch the electric field distribution.

(b) What will be the breakdown voltage of the diode if the width of the n-region is 1 μm? Sketch the electric field distribution.